



Volumes of Revolution · Worksheet 1

Core Pure 1 Chapter 5: Volumes of Revolution. Answer all questions. Show your working clearly. Total: 65 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Every rotation is through 2π radians about the stated axis. Give exact answers in terms of π unless a question says otherwise. Answers obtained only from a calculator's numerical integration will not gain full marks.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Volumes of revolution about the x -axis	20	
B Volumes of revolution about the y -axis	20	
C Volumes involving subtraction and modelling	20	
Roar-level challenge	5	
Total	65	

Section A Volumes of revolution about the x -axis

20 marks

1. The region bounded by the line $y = 3x$, the x -axis and the line $x = 2$ is rotated about the x -axis. [2]
Use integration to find the volume of the solid generated.

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2. The region bounded by the curve $y = x^2 + 1$, the coordinate axes and the line $x = 1$ is rotated about the x -axis. [3]
Find the exact volume of the solid generated.

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3. The region bounded by the curve $y = \frac{1}{x}$, the x -axis and the lines $x = 1$ and $x = 4$ is rotated about the x -axis. [3]

Find the exact volume of the solid generated.

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4. The region R is bounded by the curve $y = x(4 - x)$ and the x -axis. [4]

Find the exact volume of the solid generated when R is rotated about the x -axis.

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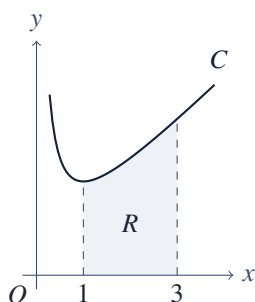
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5. The curve C has equation $y = x + \frac{1}{x}$, $x > 0$. The region R , shown shaded, is bounded by C , the x -axis and the lines $x = 1$ and $x = 3$.



- (a) Find the coordinates of the minimum point of C . [2]

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- (b) Find the exact volume of the solid generated when R is rotated about the x -axis. [4]

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- (c) Without further integration, explain why this volume is greater than 8π . [2]

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(Total for Question 5 is 8 marks)



Section B Volumes of revolution about the y -axis

20 marks

6. The region bounded by the curve $y = x^2$, $x \geq 0$, the y -axis and the line $y = 4$ is rotated about the y -axis. [2]

Find the exact volume of the solid generated.

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7. The region bounded by the curve $y = x^3$, the y -axis and the lines $y = 1$ and $y = 8$ is rotated about the y -axis. [3]

Find the exact volume of the solid generated.

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8. The region bounded by the curve $y = \ln x$, the coordinate axes and the line $y = 2$ is rotated about the y -axis. [3]

Find the exact volume of the solid generated.

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9. The curve C has equation $y = \frac{2}{x+1}$, $x \geq 0$. The region bounded by C , the y -axis and the line $y = 1$ is rotated about the y -axis. [4]

Show that the volume of the solid generated is $\pi(3 - 4\ln 2)$.

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10. The curve C has equation $x = 4 - y^2$. The region R is bounded by C and the positive coordinate axes.

(a) Find the coordinates of the points where C meets the positive x -axis and the positive y -axis. [2]

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(b) Find the exact volume of the solid generated when R is rotated about the y -axis. [4]

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(c) Find the exact volume of the solid generated when R is rotated about the x -axis. [2]

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(Total for Question 10 is 8 marks)



Section C Volumes involving subtraction and modelling

20 marks

11. The region bounded by the line $y = x$, the line $y = 2$ and the y -axis is rotated about the x -axis. [2]
Find the exact volume of the solid generated.

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12. The region enclosed between the line $y = x$ and the curve $y = x^2$ is rotated about the x -axis. [3]
Find the exact volume of the solid generated.

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13. The region enclosed between the line $y = 2x$ and the curve $y = x^2$ is rotated about the y -axis. [3]
Find the exact volume of the solid generated.

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14. The region R is bounded by the curve $y = 4 - x^2$ and the line $y = 3$. The region R is rotated about the x -axis to form a ring-shaped solid. [4]
Find the exact volume of this solid.

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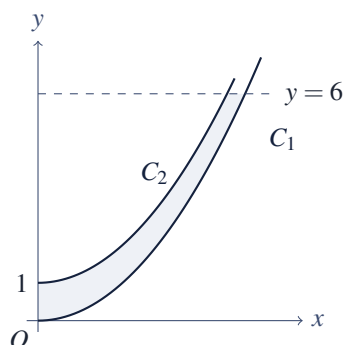
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15. A designer models a bowl. The region R , shown shaded, is bounded by the curves C_1 and C_2 and the line $y = 6$, where

$$C_1 : y = \frac{1}{5}x^2 \quad \text{and} \quad C_2 : y = \frac{1}{5}x^2 + 1, \quad x \geq 0.$$

The bowl is the solid formed when R is rotated about the y -axis. The units are centimetres.



- (a) Show that the volume of material used to make the bowl is $\frac{55\pi}{2} \text{ cm}^3$. [4]

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- (b) The bowl is placed on a table and water is poured in to a depth of 3 cm. Find the volume of water in the bowl, giving your answer to 3 significant figures. [2]

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- (c) The material has density 2.4 g cm^{-3} . Find the mass of the bowl. [1]

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- (d) State one limitation of the model. [1]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. The circle with equation $x^2 + (y - 3)^2 = 1$ is rotated through 2π radians about the x -axis to form a ring-shaped solid (a torus). [5]
By writing the upper and lower halves of the circle as $y = 3 \pm \sqrt{1 - x^2}$, show that the volume of the solid is $6\pi^2$.

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END OF WORKSHEET

Total: 65 marks